3(a)	P: total energy Q: potential energy R: kinetic energy	**B2**
3(b)	$E = ½m\omega^2x_0^2$ **or** $E = ½mv_0^2$ **and** $v_0 = \omega x_0$	**C1**
	$6.4 \times 10^{-3} = \frac{1}{2} \times 0.130 \times \omega^2 \times 0.015^2$ ($\omega^2 = 438$) ($\omega = 20.9$)	**C1**
	$T = 2\pi / \omega$	**C1**
	$= 2\pi / 20.9$ $= 0.30$ s	**A1**
3(c)(i)	resistive forces	**B1**
3(c)(ii)	0.92^6	**C1**
	decrease in energy $= 6.4 - (6.4 \times 0.92^6)$ $= 2.5$ mJ	**A1**
3(c)(iii)	light damping because the amplitude of oscillations gradually reduces **or** light damping because the system still oscillates	**B1**